

1.5 Orthogonality and orthogonal matrices

1.5.1 Orthogonal vectors and complements

Vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. For orthogonal vectors the Pythagorean identity $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ holds. A set of nonzero, mutually orthogonal vectors is independent: if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$, taking the inner product with $\mathbf{v}_k$ leaves $c_k \|\mathbf{v}_k\|^2 = 0$. A set is **orthonormal** if, in addition, each vector has unit length. The columns of an $n \times r$ matrix $\mathbf{Q}$ are orthonormal iff $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$.

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$. The next proposition is all that this chapter needs about complements. Chapter 6 builds the theory of orthogonal projections on it.

Proposition 1.5.1

Let $\mathbf{A}$ be $m \times n$ and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace.

- (a) $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ and $\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp$.
- (b) $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$, and $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$. In particular $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ and $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$.
- (c) $(\mathcal{S}^\perp)^\perp = \mathcal{S}$.

Proof.

- (a) $\mathbf{A}\mathbf{v} = \mathbf{0}$ says that $\mathbf{v}$ is orthogonal to every row of $\mathbf{A}$, hence to their span. Apply this to $\mathbf{A}^\top$ for the second identity.
- (b) Put a basis of $\mathcal{S}$ (k vectors) into the columns of $\mathbf{B}$. Then $\mathcal{S}^\perp = \mathcal{N}(\mathbf{B}^\top)$, which has dimension $n - k$ by rank-nullity. If $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$ then $\mathbf{u}^\top \mathbf{u} = 0$, so the sum is direct and has dimension n , so it is $\mathbb{R}^n$.
- (c) $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$, and both have dimension $\dim \mathcal{S}$ by (b). ■

1.5.2 Gram–Schmidt and the QR factorization

Theorem 1.5.2. Gram–Schmidt; QR factorization

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$ be $n \times p$ with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are orthonormal and, for each j , span the same subspace as $\mathbf{x}_1, \dots, \mathbf{x}_j$. Equivalently, $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}$ upper triangular with positive diagonal entries.

Proof. By induction on j . Suppose $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ are orthonormal and span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$. Then $\mathbf{w}_j \neq \mathbf{0}$, since otherwise $\mathbf{x}_j$ would lie in that span, contradicting independence. For $i < j$, $\mathbf{u}_i^\top \mathbf{w}_j = \mathbf{u}_i^\top \mathbf{x}_j - \mathbf{u}_i^\top \mathbf{x}_j = 0$. So $\mathbf{u}_1, \dots, \mathbf{u}_j$ are orthonormal, and they span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ because $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k$. That last identity, read column by column, is $\mathbf{X} = \mathbf{QR}$ with $r_{jj} = \|\mathbf{w}_j\| > 0$ and $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$ for $k < j$. ■

The factorization is unique (Exercise (B2)). Numerical software computes it by a more stable route, but the object is the same, and Chapter 6 and Chapter 10 rely on it. Two consequences are used repeatedly. First, any orthonormal set $\mathbf{u}_1, \dots, \mathbf{u}_r$ in $\mathbb{R}^n$ extends to an orthonormal basis: apply Gram–Schmidt to $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$, discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

1.5.3 Orthogonal matrices

A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. By Proposition 1.3.1, $\mathbf{Q}^\top$ is then the inverse of $\mathbf{Q}$, so also $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$: the rows of an orthogonal matrix are orthonormal as well as its columns.

Proposition 1.5.3

Let $\mathbf{Q}$ and $\mathbf{P}$ be $n \times n$ orthogonal matrices.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{v}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{v}$.
- (b) $\mathbf{Q}^\top$ and $\mathbf{Q}\mathbf{P}$ are orthogonal.
- (c) Every entry satisfies $|q_{ij}| \leq 1$.
- (d) If $[\mathbf{Q}_1, \mathbf{Q}_2]$ is orthogonal with $\mathbf{Q}_1$ of size $n \times r$, then $\mathbf{Q}_1 \mathbf{Q}_1^\top + \mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I}_n$, and $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$.

Proof.

- (a) $(\mathbf{Q}\mathbf{x})^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{v}$. (b) $(\mathbf{Q}\mathbf{P})^\top \mathbf{Q}\mathbf{P} = \mathbf{P}^\top \mathbf{P} = \mathbf{I}$.
- (b) Each column has unit length, so no entry can exceed 1 in modulus.
- (c) The first statement is $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ written in blocks. For the second, $\mathbf{Q}_1^\top \mathbf{Q}_2 = \mathbf{0}$ gives $\mathcal{C}(\mathbf{Q}_2) \subseteq \mathcal{C}(\mathbf{Q}_1)^\perp$, and both have dimension $n - r$. ■

The determinant of an orthogonal matrix is ± 1 , since $1 = \det(\mathbf{Q}^\top \mathbf{Q}) = (\det \mathbf{Q})^2$. That argument multiplies determinants, which are not defined until Section 1.6, so the claim is stated and proved there, as Proposition 1.6.2(f), rather than in part (c).

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of Chapter 4 can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

Example. Helmert matrices

The $n \times n$ **Helmert matrix** $\mathbf{H}$ has first row $n^{-1/2}\mathbf{1}^\top$ and, for $k = 2, \dots, n$, k th row

$$\mathbf{h}_k^\top = \frac{1}{\sqrt{k(k-1)}}(\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0).$$

Each row has unit length, since $(k-1) + (k-1)^2 = k(k-1)$. Row k is orthogonal to row 1 because its entries sum to zero. For $2 \leq j < k$, the nonzero entries of $\mathbf{h}_j$ sit in positions $1, \dots, j$, where $\mathbf{h}_k$ is constant, so $\mathbf{h}_j^\top \mathbf{h}_k$ is proportional to the sum of the entries of $\mathbf{h}_j$, which is zero. So $\mathbf{H}$ is orthogonal. For $\mathbf{y} \in \mathbb{R}^n$, $(\mathbf{H}\mathbf{y})_1 = \sqrt{n}\bar{y}$ and, by Proposition 1.5.3(a),

$$\sum_{k=2}^n (\mathbf{h}_k^\top \mathbf{y})^2 = \|\mathbf{H}\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of $n-1$ squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent (Exercise (B1)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector $\mathbf{u}$, the **Householder reflection** $\mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ is symmetric and orthogonal and reflects $\mathbb{R}^n$ in the hyperplane $\mathbf{u}^\perp$. Householder reflections are how QR factorizations are computed in practice (Exercise (B1)).

1.5.4 Exercises

1.5.4.1 A. Check your understanding

Exercise. A1

Show that every 2×2 orthogonal matrix is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (determinant 1) or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ (determinant -1). Find the line fixed by the reflection.

1.5.4.2 B. Practice

Exercise. B1

Let $\mathbf{u}$ be a unit vector and $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$. Show that $\mathbf{H}$ is symmetric and orthogonal, that $\mathbf{H}\mathbf{u} = -\mathbf{u}$, and that $\mathbf{H}\mathbf{v} = \mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$. Given $\mathbf{x}$ not a nonnegative multiple of $\mathbf{e}_1$, find $\mathbf{u}$ with $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$. Explain how repeating this column by column produces a QR factorization.

Solution. $\mathbf{H}^\top = \mathbf{H}$, and $\mathbf{H}^\top \mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^\top + 4\mathbf{u}(\mathbf{u}^\top \mathbf{u})\mathbf{u}^\top = \mathbf{I}$. Also $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$, and $\mathbf{H}\mathbf{v} = \mathbf{v}$ when $\mathbf{u}^\top \mathbf{v} = 0$. Let $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$ and $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. Then $\mathbf{w}^\top \mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^\top \mathbf{x}$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top \mathbf{x})/(\mathbf{w}^\top \mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$. Applying such a reflection $\mathbf{H}_1$ to the first column of $\mathbf{X}$, then a reflection acting on coordinates $2, \dots, n$ to the second column of $\mathbf{H}_1\mathbf{X}$, and so on, gives $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{R}$ upper triangular. Hence $\mathbf{X} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ holds the first p columns of the orthogonal matrix $\mathbf{H}_1 \cdots \mathbf{H}_p$. In floating point one uses $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ instead, which avoids cancellation. It gives

$\mathbf{H}\mathbf{x} = -\operatorname{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$, so diagonal entries of $\mathbf{R}$ may come out negative; changing the sign of such a row of $\mathbf{R}$ and of the matching column of $\mathbf{Q}$ at the end restores the positive diagonal of Theorem 1.5.2.

Exercise. B2

Show that the QR factorization of a matrix with full column rank, with $\mathbf{R}$ having positive diagonal, is unique, and that $\mathbf{R}^\top$ is the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$.

Solution. If $\mathbf{X} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$, then $\mathbf{X}^\top\mathbf{X} = \mathbf{R}_1^\top\mathbf{R}_1 = \mathbf{R}_2^\top\mathbf{R}_2$. Each $\mathbf{R}_k^\top$ is lower triangular with positive diagonal, so both are the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$, which is unique by Theorem 1.7.4(d). So $\mathbf{R}_1 = \mathbf{R}_2$ and $\mathbf{Q}_1 = \mathbf{X}\mathbf{R}_1^{-1} = \mathbf{Q}_2$.

1.5.4.3 C. Going deeper

Exercise. C1

Let $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_k]$ and $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_k]$ be $n \times k$ matrices with the same cross-product matrix, $\mathbf{A}^\top\mathbf{A} = \mathbf{B}^\top\mathbf{B}$. Show that there is an orthogonal $n \times n$ matrix $\mathbf{Q}$ with $\mathbf{B} = \mathbf{Q}\mathbf{A}$. So a Gram matrix determines its vectors up to a rotation or reflection of $\mathbb{R}^n$, and two model matrices with the same $\mathbf{X}^\top\mathbf{X}$ describe the same geometry in different coordinates. *Hint:* show first that $\mathbf{A}\mathbf{c} = \mathbf{0}$ iff $\mathbf{B}\mathbf{c} = \mathbf{0}$.

Solution. For every $\mathbf{c} \in \mathbb{R}^k$, $\|\mathbf{A}\mathbf{c}\|^2 = \mathbf{c}^\top\mathbf{A}^\top\mathbf{A}\mathbf{c} = \mathbf{c}^\top\mathbf{B}^\top\mathbf{B}\mathbf{c} = \|\mathbf{B}\mathbf{c}\|^2$, so $\mathbf{A}\mathbf{c} = \mathbf{0}$ iff $\mathbf{B}\mathbf{c} = \mathbf{0}$. Hence the rule $\mathbf{T}(\mathbf{A}\mathbf{c}) = \mathbf{B}\mathbf{c}$ defines a map on $\mathcal{C}(\mathbf{A})$ without ambiguity: if $\mathbf{A}\mathbf{c} = \mathbf{A}\mathbf{d}$, then $\mathbf{A}(\mathbf{c} - \mathbf{d}) = \mathbf{0}$, so $\mathbf{B}(\mathbf{c} - \mathbf{d}) = \mathbf{0}$. The map is linear and preserves lengths, hence also inner products (Exercise (A2)). Both $\mathcal{C}(\mathbf{A})$ and $\mathcal{C}(\mathbf{B})$ have dimension $r = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{B})$, the two ranks being equal because the null spaces agree (Theorem 1.2.2).

Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be an orthonormal basis of $\mathcal{C}(\mathbf{A})$, which exists by Theorem 1.5.2. Then $\mathbf{T}\mathbf{u}_1, \dots, \mathbf{T}\mathbf{u}_r$ are orthonormal and lie in $\mathcal{C}(\mathbf{B})$, so they are an orthonormal basis of it. Extend both sets to orthonormal bases $\mathbf{u}_1, \dots, \mathbf{u}_n$ and $\mathbf{w}_1, \dots, \mathbf{w}_n$ of $\mathbb{R}^n$, with $\mathbf{w}_i = \mathbf{T}\mathbf{u}_i$ for $i \leq r$, and put $\mathbf{Q} = \mathbf{W}\mathbf{U}^\top$, where $\mathbf{U}$ and $\mathbf{W}$ hold these bases as columns. Both are orthogonal, so $\mathbf{Q}$ is orthogonal by Proposition 1.5.3(b), and $\mathbf{Q}\mathbf{u}_i = \mathbf{w}_i$ for every i . For $\mathbf{c} \in \mathbb{R}^k$, write $\mathbf{A}\mathbf{c} = \sum_{i \leq r} \alpha_i \mathbf{u}_i$; then $\mathbf{Q}\mathbf{A}\mathbf{c} = \sum_{i \leq r} \alpha_i \mathbf{T}\mathbf{u}_i = \mathbf{T}(\mathbf{A}\mathbf{c}) = \mathbf{B}\mathbf{c}$. Taking $\mathbf{c} = \mathbf{e}_j$ gives $\mathbf{Q}\mathbf{a}_j = \mathbf{b}_j$ for each j , that is, $\mathbf{Q}\mathbf{A} = \mathbf{B}$.

Exercise. C2

Show that every $n \times n$ orthogonal matrix is a product of at most n Householder reflections. *Hint:* for unit vectors $\mathbf{x} \neq \mathbf{y}$, the reflection with $\mathbf{u} = (\mathbf{x} - \mathbf{y})/\|\mathbf{x} - \mathbf{y}\|$ sends $\mathbf{x}$ to $\mathbf{y}$; use it on the first column and induct.

Solution. For unit $\mathbf{x} \neq \mathbf{y}$ put $\mathbf{w} = \mathbf{x} - \mathbf{y}$, so $\mathbf{w}^\top\mathbf{w} = 2 - 2\mathbf{x}^\top\mathbf{y}$ and $\mathbf{w}^\top\mathbf{x} = 1 - \mathbf{y}^\top\mathbf{x} = \frac{1}{2}\mathbf{w}^\top\mathbf{w}$. With $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$ and $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ (Exercise (B1)), $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top\mathbf{x})/(\mathbf{w}^\top\mathbf{w}) = \mathbf{x} - \mathbf{w} =$

$\mathbf{y}$.

Argue by induction on n , the case $n = 1$ being $\mathbf{Q} = (\pm 1)$, where $\mathbf{Q} = (1)$ is the empty product and $\mathbf{Q} = (-1)$ is the reflection with $\mathbf{u} = (1)$. Let $\mathbf{Q}$ be $n \times n$ orthogonal with first column $\mathbf{q}_1$, a unit vector. If $\mathbf{q}_1 = \mathbf{e}_1$, take $\mathbf{H}_1 = \mathbf{I}$ (no reflection); otherwise take the reflection $\mathbf{H}_1$ above with $\mathbf{x} = \mathbf{q}_1$ and $\mathbf{y} = \mathbf{e}_1$. Then $\mathbf{H}_1\mathbf{Q}$ is orthogonal with first column $\mathbf{e}_1$, and since its rows are orthonormal too, its first row is $\mathbf{e}_1^\top$. So $\mathbf{H}_1\mathbf{Q} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}_2 \end{pmatrix}$ with $\mathbf{Q}_2$ orthogonal of order $n - 1$. By induction $\mathbf{Q}_2$ is a product of at most $n - 1$ reflections $\mathbf{I}_{n-1} - 2\mathbf{v}_j\mathbf{v}_j^\top$, and $\begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{I}_{n-1} - 2\mathbf{v}_j\mathbf{v}_j^\top \end{pmatrix} = \mathbf{I}_n - 2\tilde{\mathbf{v}}_j\tilde{\mathbf{v}}_j^\top$ with $\tilde{\mathbf{v}}_j = (0, \mathbf{v}_j^\top)^\top$ a unit vector. Hence $\mathbf{H}_1\mathbf{Q}$ is a product of at most $n - 1$ reflections, and $\mathbf{Q} = \mathbf{H}_1(\mathbf{H}_1\mathbf{Q})$ of at most n , because $\mathbf{H}_1^{-1} = \mathbf{H}_1$.

Exercise. C3

Let $\mathbf{X} = \mathbf{QR}$ be the QR factorization of an $n \times p$ matrix of full column rank. Show that $\det(\mathbf{X}^\top \mathbf{X}) = \prod_j r_{jj}^2$ and that $r_{jj} \leq \|\mathbf{x}_j\|$, with equality iff $\mathbf{x}_j$ is orthogonal to $\mathbf{x}_1, \dots, \mathbf{x}_{j-1}$. Deduce **Hadamard's inequality** $\det(\mathbf{X}^\top \mathbf{X}) \leq \prod_j \|\mathbf{x}_j\|^2$, with equality iff the columns are mutually orthogonal, and $|\det \mathbf{A}| \leq \prod_j \|\mathbf{a}_j\|$ for a square matrix $\mathbf{A}$. (Determinants are in [Section 1.6](#).) The quantity $\det(\mathbf{X}^\top \mathbf{X})$ is a squared volume: it is largest, for columns of given lengths, when the regressors are orthogonal.

Solution. From $\mathbf{X} = \mathbf{QR}$ with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ we get $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$, so by Proposition 1.6.2(a), (c) and (d), $\det(\mathbf{X}^\top \mathbf{X}) = (\det \mathbf{R})^2 = \prod_j r_{jj}^2$. The j th column of $\mathbf{X} = \mathbf{QR}$ is $\mathbf{x}_j = \sum_{k \leq j} r_{kj} \mathbf{u}_k$, a combination of orthonormal vectors, so $\|\mathbf{x}_j\|^2 = \sum_{k \leq j} r_{kj}^2 \geq r_{jj}^2$, with equality iff $r_{kj} = 0$ for every $k < j$. Since $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$ and $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$ by [Theorem 1.5.2](#), that happens iff $\mathbf{x}_j$ is orthogonal to the earlier columns. Multiplying the p inequalities gives Hadamard's inequality, with equality iff every column is orthogonal to all its predecessors, that is, iff the columns are mutually orthogonal. For a square $\mathbf{A}$: if it is singular both sides of $(\det \mathbf{A})^2 = \det(\mathbf{A}^\top \mathbf{A}) \leq \prod_j \|\mathbf{a}_j\|^2$ are handled by $\det \mathbf{A} = 0$, and otherwise the inequality just proved applies with $\mathbf{X} = \mathbf{A}$.